

Complex Network Analysis of Social Relationships and Personality from TV Drama Dialogues

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Abstract: Dialogues are linguistic interactions between people and can provide hints on social relationship and personality. We aim to analyze the social relationships of the characters in TV dramas based on dialogues. In addition to knowing just who talks to whom, the analysis of dialogue content gives more detailed information on the types of interaction, e.g. asking or informing, and the emotional status. We use complex network measures, such as path lengths, geodesics, cluster coefficient, and centrality, to analyze the structural and functional properties of the dialogue network. Among others, we found the betweenness centrality measure informative for identifying the social relationships since this assesses the importance of a node in a complex network in which small worldness is inherent. Analyzing the complete corpus of TV drama Friends aired for 10 years, we discovered the connector nodes which are essentially the social equivalent of a hub of computer network. Complexity measures of the dialogue network also help characterize the personality of the main characters in the dramas we studied. The text-based analysis of the present work can be extended to incorporate the sound modality to capture further information on the characters and their interactions.